

INVITED REVIEW

Bioengineering Approaches for Nanomedicine Delivery in Therapy of Myocardial Infarction

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ABSTRACT: Cardiovascular disease remains the leading cause of death globally. In particular, ischemic heart disease represents the most common condition in clinics. Adult hearts have minimal regenerative capacity. Upon myocardial infarction, the formation of activated myofibroblasts replaces the massive loss of cardiomyocytes. Nanomedicine has emerged as a promising strategy for cardiovascular disease treatment, with synthetic nanoparticles and extracellular vesicles serving as representative delivery platforms. However, directly applying these nanomedicines faces several challenges, including insufficient cardiac targeting, poor retention rate at the injured site, and potential biocompatibility concerns. To overcome these limitations, different bioengineering approaches have been adopted. In this review, we discussed the landscape of cardiovascular nanomedicine, encompassing both synthetic nanocarriers and biological carriers and targeted delivery. We further evaluated the engineering approaches to enhance carrier performance. Multiple administration routes are also compared. Collectively, these advances represent a shift toward an integrated, multifunctional delivery platform with the potential to translate cardiac nanomedicine from bench to bedside.

Key Words: blood pressure ■ cardiovascular diseases ■ drug liberation ■ heart rate ■ lipid nanoparticles

Cardiovascular diseases (CVDs) are the leading cause of death worldwide. According to the 2025 Heart Disease and Stroke Statistics Update from the American Heart Association, CVD accounts for ≈ 19.41 million deaths worldwide in 2021 and 941 652 deaths recorded in 2022 in the United States.¹ Among CVDs, ischemic heart disease and myocardial infarction (MI) represent the major drivers of morbidity.² The adult mammalian heart has negligible regenerative capability, with cardiomyocyte regeneration being $<1\%$ per year.^{3,4} Consequently, the massive loss of cardiomyocytes following the ischemic event is replaced by the formation of fibrotic scar tissue, leading to reduced cardiac contractility and eventual heart failure.

Current medications, such as ACE (angiotensin-converting enzyme) inhibitors and β -blockers, are used to reduce cardiac workload by lowering blood pressure and heart rate.⁵ These drugs only palliatively treat the disease, with potential side effects caused by non-specific tissue distribution and requiring clinical monitoring. Over the past few decades, researchers have developed nanomedicine-based delivery systems that offer targeted therapeutic approaches and enhance

regenerative potential.⁶ However, although these nanomedicine approaches showed great potential in vitro, many failed during clinical trials due to the following barriers.^{7–9} First, the retention rate in the heart is low. The heart is a contractile pump. It exerts a washout effect that can rapidly clear the material injected in the myocardium, resulting in $<10\%$ retention rates after administration.^{10,11} Second, the nanomedicines delivered systemically often experience poor accumulation at the target cardiac site. Over 90% of the therapeutics are sequestered by the reticuloendothelial system, hindering from reaching the cardiac tissue.¹² Beyond the nonspecific accumulation and low retention/accumulation, the postinfarct microenvironment presents additional challenges: acidic pH,¹³ reactive oxygen species (ROS),¹⁴ and inflammatory infiltration. This microenvironment can degrade the nanomedicine at the target site, as well as the cargo therapeutics inside the particles. However, the same signals can also be exploited as endogenous triggers for on-demand drug release. For example, Li et al cross-linked ROS-sensitive poly(vinyl alcohol) hydrogel with a thioborate linker, which was injected intrapericardially to deliver basic fibroblast growth factor, releasing

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Nonstandard Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
ApoE	apolipoprotein E
CHP	cardiac homing peptide
CVD	cardiovascular disease
EV	extracellular vesicle
FAP	fibroblast activation protein
LDLR	low-density lipoprotein receptor
LNP	lipid nanoparticle
MI	myocardial infarction
MIF	macrophage migration inhibitory factor
MSC	mesenchymal stem cell
NF-κB	nuclear factor- κ B
PLGA	poly(lactic-co-glycolic acid)
ROS	reactive oxygen species
TLR	toll-like receptor
VSV-G	vesicular stomatitis virus G glycoprotein

cargo in response to ROS overproduction in the infarcted myocardium and improving cardiac function in a rat I/R model.¹⁴ Yang et al¹⁵ used ROS-responsive poly(vinyl alcohol)-TSPBA hydrogel matrix to load hollow mesoporous polydopamine nanoparticles. The nanoparticles are filled with mannose-functionalized Danshensu sodium for promoting endothelial tube formation while modulating inflammation after myocardial infarction.¹⁵

To overcome these hurdles, several bioengineering approaches have been developed. Lipid nanoparticles (LNPs), for instance, are clinically validated nonviral vectors capable of delivering small molecules, as well as mRNA and siRNA therapeutics.^{16,17} Despite their translational success, conventional LNPs lack inherent cardiac tropism and are often limited by substantial off-target accumulation in the liver. In contrast, extracellular vesicles (EVs), the natural nanoparticles secreted by all types of cells, exhibit superior biocompatibility and low immunogenicity, yet their clinical scalability remains challenged by limited production yields and rapid systemic clearance. In this review, we evaluate various emerging bioengineering strategies to circumvent physiological barriers in drug delivery for CVDs. We further compare distinct administration routes and discuss how route-specific considerations can be leveraged to enhance cardiac precision and therapeutic efficacy.

THE LANDSCAPE OF CARDIOVASCULAR NANOMEDICINE

Nanocarriers for drug delivery in CVDs can be divided into 2 classes based on their composition and origin: synthetic and biological nanocarriers. Synthetic nanocarriers are engineered from chemical precursors, where the

Highlights

- Cardiovascular disease, especially myocardial infarction, remains the leading cause of death globally, and the adult heart has minimal regenerative capacity, driving the need for targeted nanomedicine delivery strategies.
- Both synthetic nanocarriers and biological extracellular vesicles offer distinct advantages for cardiac drug delivery. However, both strategies face challenges of low cardiac retention and off-target accumulation.
- Bioengineering strategies, such as parental cell preconditioning and surface modification with cardiac homing peptides, improve extracellular vesicle targeting efficiency and therapeutic potency for cardiac repair.
- Extrinsic surface modification and intrinsic lipid engineering improve lipid nanoparticle and nanoparticle biodistribution toward the heart.
- Local delivery routes achieve higher cardiac retention than systemic injection. Hydrogel incorporation can further improve cardiac retention and achieve controlled, sustained release.

size, surface charge, and degradation mechanism can be precisely controlled. On the contrary, biological carriers are naturally secreted by cells and possess inherent biocompatibility with complex surface markers for better intercellular interaction. The summary of nanocarriers discussed is shown in Figure 1.

Synthetic Nanocarriers

Synthetic nanocarriers have long been the cornerstone of drug delivery, evolving from simple spheres (eg, poly(lactic-co-glycolic acid; PLGA)) to complex lipid-based systems. LNPs are currently a promising vector for drug delivery.^{18,19} LNPs are composed of a mixture of 4 components: ionizable lipid, cholesterol, helper lipids, and PEG lipid.²⁰ The ionizable lipids allow the LNP to remain neutral at physiological pH but become positively charged in the acidic endosome (pH $\approx$ 5–6), facilitating the release of cargo into the cytoplasm.²¹ LNPs were used in the first US Food and Drug Administration–approved siRNA-based therapy, Patisiran (Onpattro, Alnylam, Inc), in 2018 for the treatment of amyloidosis in adult patients.²² The clinical utility of LNPs has been further demonstrated by the development of COVID-19 mRNA vaccines, Comirnaty (Pfizer-BioNTech) and Spikevax (Moderna).^{23,24} However, their application in CVDs is limited. ApoE (apolipoprotein E) is a component of plasma lipoprotein that can bind to several ApoE receptors, such as the LDLR (low-density lipoprotein receptor). Intravenously injected LNPs can bind to ApoE, resulting in targeting primarily to the liver via LDLR, a

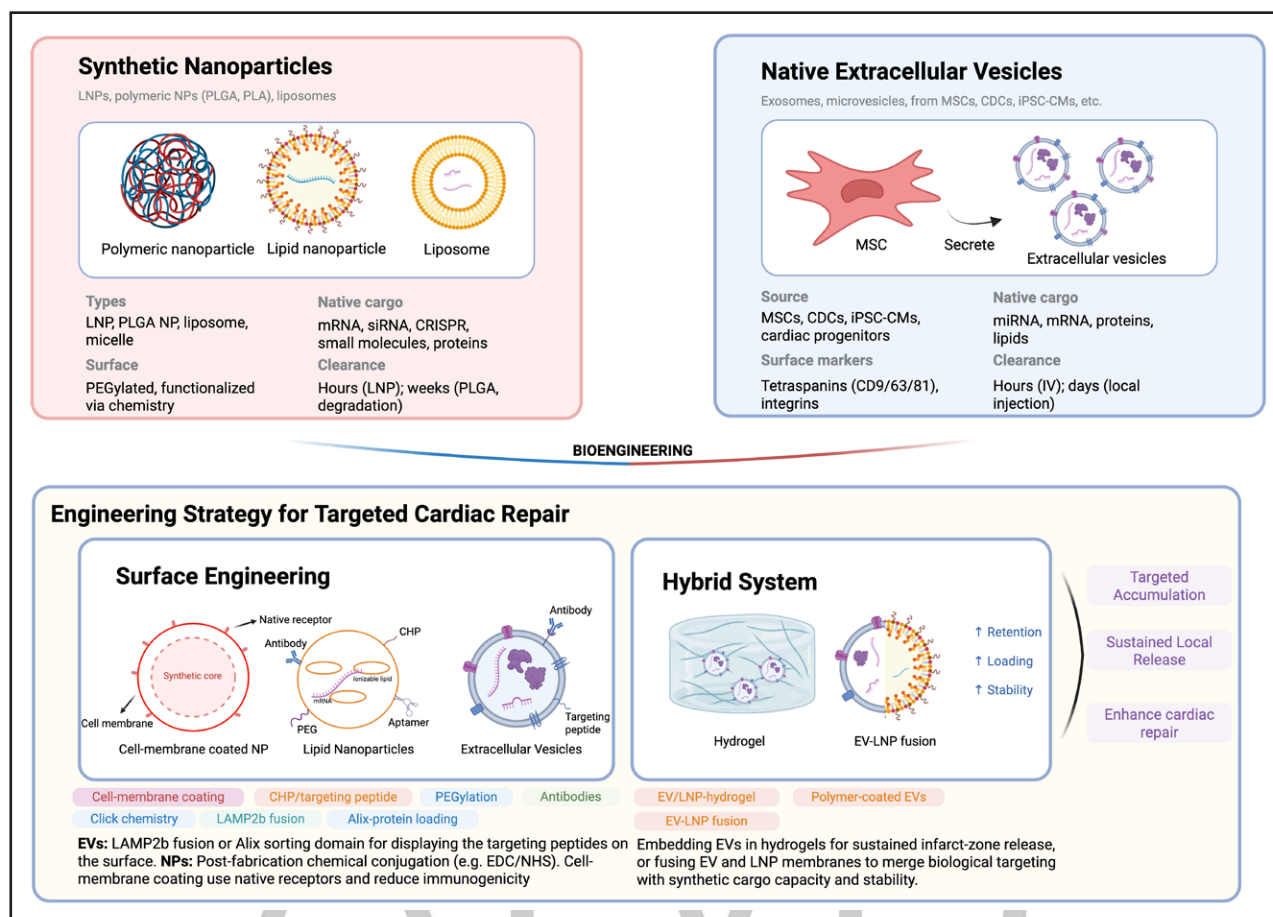


Figure 1. Bioengineering strategies for nanocarrier-based cardiac drug delivery for Cardiovascular diseases.

Top, Native extracellular vesicles (EVs) with innate cardioprotective cargo and synthetic nanoparticles (NPs, eg, lipid nanoparticles [LNPs] and poly(lactic-co-glycolic acid); PLGA) with tunable properties. **Bottom,** Engineering strategies. Surface engineering: cell membrane-coated NPs leverage native receptors and improve biocompatibility of synthetic nanoparticles or EVs; LNPs are functionalized with CHP (cardiac homing peptide), antibodies, and PEG; and EVs display cardiac targeting properties via LAMP2b (lysosome-associated membrane protein 2b) fusion or Alix (ALG-2-interacting protein X) sorting domain-mediated loading on the surface. Hybrid systems: EV-composite or patches provide sustained release in the targeted infarct zone and provide mechanical support to the epicardium. EV-LNP membrane fusion merges biological targeting with synthetic cargo loading capacity. The engineered carriers achieved targeted accumulation, sustained local release, and enhanced repair outcomes. CDC indicates cardiosphere-derived cell; iPSC-CM, induced pluripotent stem cell-cardiomyocyte; MSC, mesenchymal stem cell; and PEG, polyethylene glycol.

tropism that is highly dependent on lipid composition, surface chemistry, and route of administration rather than an inherent property of all LNP formulations.²⁵ This hepatic trap results in <1% to 2% of the injected dose accumulating in the heart while enhancing the burden on organs such as the liver. PLGA nanoparticles are another synthetic option with a long track record. The polymer breaks down through hydrolysis into biocompatible byproducts, offering tunable drug release over days to weeks depending on polymer molecular weight and composition. Compared with LNPs, PLGA particles handle a broader range of payloads, both water-soluble and hydrophobic drugs, and tend to hold up better under physiological conditions. The stability makes them a natural fit for sustained local delivery as seen in their use in cardiac patches and microneedle systems discussed later in this review.

Biological Nanocarriers

In parallel, EVs have emerged as a biological alternative to synthetic vectors. EVs are lipid-bilayer vesicles secreted by cells to transport molecules and to facilitate intercellular interaction. EVs are naturally equipped with multiple surface proteins (eg, tetraspanins and integrins) that contribute to biocompatibility and enhance cellular uptake.^{26,27} Virtually, all cell types secrete EVs, and their therapeutic impact on CVDs is intrinsically dictated by their specific cellular origin. For example, EVs derived from stem cells have shown intrinsic regenerative properties.^{28,29} Mesenchymal stem cell (MSC)-derived EVs are one of the most widely studied subtypes and are promising for promoting angiogenesis, reducing myocardial apoptosis, and suppressing fibrosis.³⁰ Other sources of EVs, such as immune cells, plants, and microorganisms,

have also been used in treating CVDs.^{31–33} Despite their biocompatibility and therapeutic potential, EVs are heterogeneous; they can originate from distinct subcellular pathways (endosomal exosomes and plasma membrane-derived microvesicles) and are broadly classified into small EVs/exosomes (≈ 30 – 150 nm) of endosomal origin and microvesicles (≈ 100 – 1000 nm) shed from the plasma membrane, with larger apoptotic bodies reaching several micrometers. EVs carry heterogeneous cargo based on the physiological state of the cell.³⁴ As a result, the production of EVs can be difficult to consistently scale up, hindering their clinical translation. One practical approach to overcome low yield is membrane extrusion, in which cells are mechanically extruded through nanopore filters to generate EV-mimetic nanovesicles at yields more than 20-fold higher than naturally secreted EVs, with comparable therapeutic efficacy in cardiac repair.¹¹

ENGINEERED EVs

Native EVs, especially those from stem cells, intrinsically carry regenerative or immunomodulating signals, mostly miRNAs or proteins, and have been widely studied in the treatment of CVDs. MSC-EVs are proven to modulate inflammation, induce the formation of regulatory T cells, and promote angiogenesis by releasing miR-181b or miR-21-5p to the injured site.^{35–37} Cardiac progenitor cell-derived EVs can alleviate arrhythmogenic cardiomyopathy and reduce inflammation.³⁸ Induced pluripotent stem cell-derived EVs, particularly those derived from

induced cardiomyocytes, promote autophagy and have shown the potential to reprogram fibroblasts into functional cardiomyocytes.^{39,40} However, as mentioned previously, native EVs experience low yield, poor targeting and retention (liver trap), and insufficient cargo loading. To overcome these limitations, several bioengineering strategies have been developed, which can be classified into 3 categories: parental cell engineering, post-isolation surface modification, and cargo loading optimization. Examples are shown in Figure 2.

A common approach to modulating the composition of EVs is gene editing of the parental cell. By engineering the parental cells to overexpress specific therapeutic proteins or miRNAs, the target cargo can be significantly enriched in the secreted EVs. For instance, miR-133a-3p functions as a cardioprotective regulator by targeting the AKT signaling pathway. Researchers genetically engineered umbilical cord MSCs to overexpress the MIF (macrophage migration inhibitory factor), which enriched the miR-133a-3p content in the secreted exosomes. These engineered exosomes demonstrated superior cardioprotective function in acute MI by reducing cardiomyocyte apoptosis and promoting angiogenesis.⁴⁴ Besides genetic modification, the physiological state of the cells can also be modulated by applying environmental stimuli, such as hypoxia-preconditioning or heat shock. Cells are cultured under low oxygen levels that mimic the ischemic condition in vivo. Hypoxia preconditioning leads to the overexpression of certain RNAs or proteins that strengthen the tolerance and adaptability of cells to oxidative stress and anoxic environments. Recent

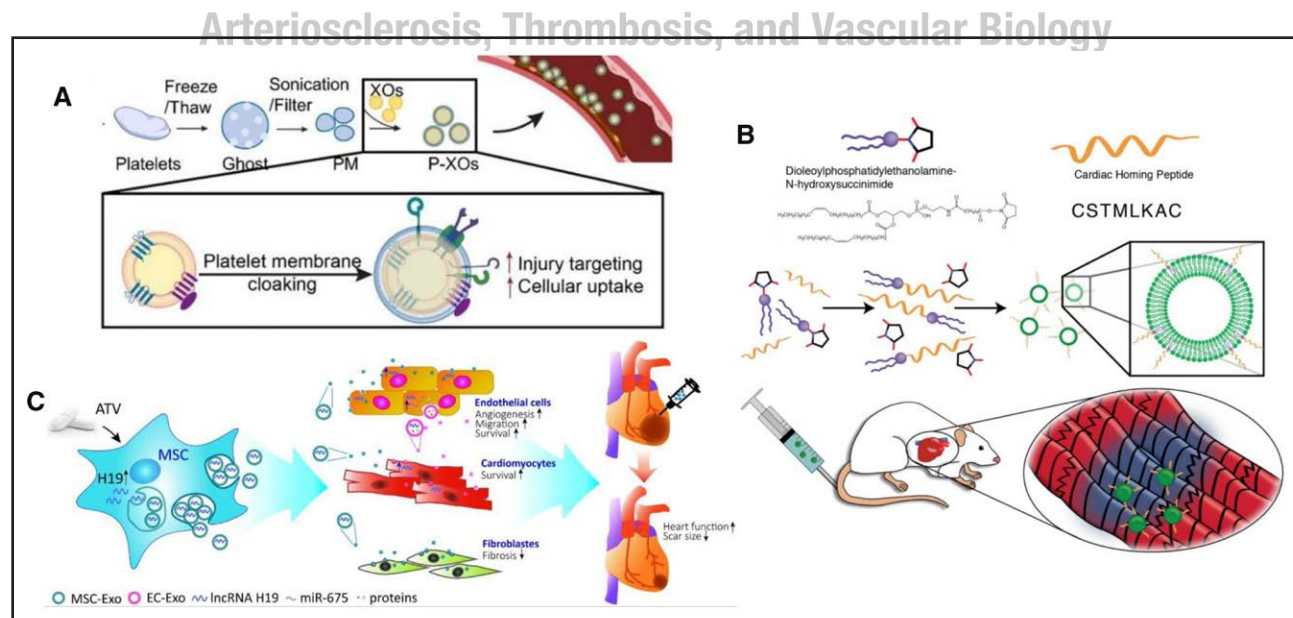


Figure 2. Bioengineering strategies to enhance targeting and therapeutic efficacy of extracellular vesicles (EVs).

A, Membrane hybridization. Platelet membrane fused with exosome to leverage native platelet receptors to enhance cellular uptake and targeted accumulation at the injury site. Adapted from Hu et al.⁴¹ **B**, Post-isolation surface modification. Cardiac homing peptides are conjugated to the exosome surface using a lipid linker to actively guide intravenously injected exosomes to the injured site. Adapted from Vandergriff et al.⁴² **C**, Parental cell preconditioning. Pretreatment of mesenchymal stem cells (MSCs) with ATV upregulates specific therapeutic cargo within the secreted MSC exosomes, which promote endothelial angiogenesis and restore heart function. Adapted from Huang et al.⁴³

studies have shown that hypoxia-preconditioned EVs show enrichment in miR-199b-3p. This miRNA targets Fgl2 to suppress the TLR (toll-like receptor) 4/NF- κ B (nuclear factor- κ B) pathway and subsequently reduce macrophage M1 polarization, alleviating myocardial inflammation.⁴⁵ Hypoxia-preconditioned MSC-EVs also promote angiogenesis and attenuate myocardial injury via upregulation of miR-214 and miR-224-5p.^{46,47} Pharmacological preconditioning, in addition to physical stress, is also used to enhance the potency of EVs. For example, atorvastatin is a small molecule that is used to treat high cholesterol and prevent heart attacks. Pretreating MSCs with atorvastatin has been proven to increase the level of long noncoding RNA H19 and miR-139-3p, promoting activation of angiogenesis and macrophage polarization, respectively.⁴³

Native EVs lack specific homing receptors for cardiac tissue and can be rapidly cleared by the reticuloendothelial system (primarily the liver and spleen). To endow EVs with active targeting capabilities, several surface modification approaches have been explored.⁴⁸ The first approach is the chemical conjugation of antibodies or CHPs (cardiac homing peptides) to the surface of the EVs. CHPs can be directly attached to the EV surface by bio-orthogonal copper-free click chemistry. Results showed that these CHP-modified EVs demonstrated a more than 2-fold increase in siRNA delivery efficiency compared with the control group.⁴⁹ However, direct chemical modification carries the risk of damaging surface proteins. An alternative approach is to insert a phospholipid tail into the lipid bilayer of EVs, leaving a hook for attaching the ligand. For example, Vandergriff et al⁴² conjugated the CHP (CSTSLKAC) to the surface of cardiosphere-derived stem cell exosomes using a dioleoylphosphatidylethanolamine N-hydroxysuccinimide lipid linker. This surface modification significantly improved the targeting of intravenously injected exosomes to the infarcted heart.^{30,42} Another approach is membrane hybridization. For example, platelet membranes are naturally embedded with multiple glycoprotein integrins and immunomodulatory proteins and possess intrinsic targeting ability toward vascular injury and inflammation. They also display markers, such as CD47 and CD55, to avoid clearance by the reticuloendothelial system. Researchers have fused platelet membranes with exosome membranes to enhance cellular uptake and promote targeting to the injury site, resulting in significantly improved cardiac function compared with unmodified exosomes.⁴¹

Once internalized, EVs face an additional barrier: endosomal entrapment. A significant fraction of internalized EVs may be sequestered in lysosomes before releasing their cargo into the cytosol. Engineering EVs with fusogenic proteins, such as VSV-G (vesicular stomatitis virus G glycoprotein), has shown promising potential in endosomal escape and improvement in intracellular cargo

delivery. For example, Liang et al⁵⁰ incorporated VSV-G into engineered EV alongside a self-cleaving domain, achieving efficient cytosolic delivery of Cre recombinase and Cas0-RNP complexes in vitro and in vivo.

For cargo loading in EVs, simple diffusion or passive incubation is generally inefficient, especially for hydrophilic molecules such as RNA. To achieve high encapsulation efficiency without affecting the biological activity of EVs, researchers have applied either physical disruption or chemical permeabilization to temporarily breach the EV membrane. Examples of physical disruption of the membrane include electroporation, sonication, and the freeze-thaw cycle. Electroporation is frequently used for encapsulating nucleic acids, specifically miRNAs and siRNAs, and small molecules. The transient pores created in the EV membrane under high voltage allow therapeutic molecules to enter the EVs. In EV-mediated cardiac therapy, electroporation is used to encapsulate miRNA such as miR-21a into MSC-EVs. However, it is reported that RNA cargo stability can be compromised during these physical loading procedures, as the high-voltage pulses of electroporation and mechanical shear of sonication may fragment RNA molecules, potentially reducing therapeutic potency.⁵¹ Tuning parameters such as voltage intensity should, therefore, be done carefully to increase loading efficiency and minimize damage to EVs.⁵² Alternatively, chemical permeabilization involves applying a surfactant such as saponin to the EV membrane. Saponin binds to the cholesterol of the EV membrane and creates pores that allow small molecules or RNA to penetrate while preserving the characteristics of the EVs.⁵³ The success of these active loading strategies relies on the balance between maximizing the encapsulation efficiency and preserving the structural and functional integrity of EVs to ensure the therapeutic efficacy in cardiac repair. Among these methods, passive incubation is gentle but inefficient for hydrophilic cargo; electroporation achieves high nucleic acid loading but risks siRNA aggregation and EV disruption; saponin permeabilization better preserves vesicle integrity while allowing RNA and small molecule loading; and parental cell engineering produces preloaded EVs without post-isolation manipulation but requires upstream genetic modification.³⁰ The optimal strategy depends on cargo type and the degree to which EV integrity must be maintained.

BIOMIMETIC NANOPARTICLES

As discussed in the first section, synthetic nanocarriers, such as PLGA nanoparticles and LNPs, have been widely explored for CVD drug delivery. PLGA nanoparticles offer sustained release and biocompatibility, while LNPs have emerged as the leading nonviral platform for siRNA and mRNA delivery. The efficacy of LNPs has been shown by their clinical success in COVID-19 mRNA vaccines and the siRNA therapeutic patisiran. However, conventional

LNPs delivered intravenously are rapidly sequestered by the liver through ApoE-mediated uptake via LDLR, resulting in <1% to 2% of the therapeutics reaching the heart. To overcome these limitations, several bioengineering strategies have been developed, which can be classified into 2 categories: extrinsic surface modification and intrinsic lipid engineering. Examples are shown in Figure 3.

Extrinsic surface modification is to chemically attach the ligand to the nanoparticle surfaces to redirect biodistribution toward cardiac tissue. Various types of cells, such as cardiomyocytes and fibroblasts, play different roles in pathological remodeling in the post-MI heart. Depending on the target cell types, these extrinsic ligands are broadly categorized into antibodies, peptides, and aptamers. Antibody-based targeting exploits the high specificity of antigen-antibody recognition. For example, Rurik et al⁵⁷ conjugated anti-CD5 antibodies onto LNP surfaces to deliver FAP (fibroblast activation protein)-encoding CAR mRNA selectively to T cells. The T cells, as a result, transiently expressed the anti-FAP CAR, turning into killer cells to recognize and eliminate FAP-positive activated cardiac fibroblasts. The results showed reduced fibrosis and restored cardiac function in a mouse model.⁵⁷ While antibodies greatly improved the targeting specificity, the density and the large molecular weight of the antibody may lead to steric hindrance on the nanoparticle surfaces, increasing immunogenicity and reducing the accessibility of binding sites.⁵⁸ Despite these advances, approaches that directly target cardiomyocytes or cardiac endothelial cells remain limited, and most strategies rely on passive accumulation at the injury site or indirect cell-mediated delivery. Peptide modification offers a more compact alternative. As discussed in the second section, CHPs have been

successfully applied to EVs; the same principle applies to synthetic nanocarriers. Weng et al⁵⁵ developed inhalable calcium phosphate nanoparticles, an inorganic carrier known for its biocompatibility and nucleic acid binding capacity, modified with CHP via DSPE-PEG linkage. This design is used for noninvasive delivery of TP-10 to the heart for pressure overload-induced heart failure.⁵⁵ Beyond ligand conjugation, coating synthetic nanoparticle cores with cell membranes provides a biomimetic targeting approach for delivery.^{54,59} Tang et al⁵⁴ developed a synthetic cell-mimicking microparticle by coating PLGA microparticles with cardiac stem cell membranes to recapitulate cell surface antigens and the therapeutic secretome. Results demonstrated therapeutic efficacy without stimulating T-cell infiltration, achieving cardiac repair without live cell transplantation.⁵⁴ Despite the versatility of these extrinsic strategies, they generally require additional conjugation or coating processes that may reduce consistency and scalability. Furthermore, the optimal density should be carefully determined to avoid impairing nanoparticle stability and cellular uptake.

An alternative to surface functionalization is to redesign the lipid composition itself to alter organ tropism, a more scalable approach that avoids additional conjugation processes. The foundational work in this area is the selective organ targeting strategy developed by the Siegwart group,⁶⁰ in which a supplemental lipid component is added to the conventional 4-component LNPs to systemically redirect biodistribution. By varying the identifying molar percentages of the selective organ targeting molecule, permanently cationic lipids (eg, DOTAP) shifted mRNA expression from the liver to the lung, while anionic lipids (eg, 19PA) redirected delivery to the spleen. While the selective organ targeting

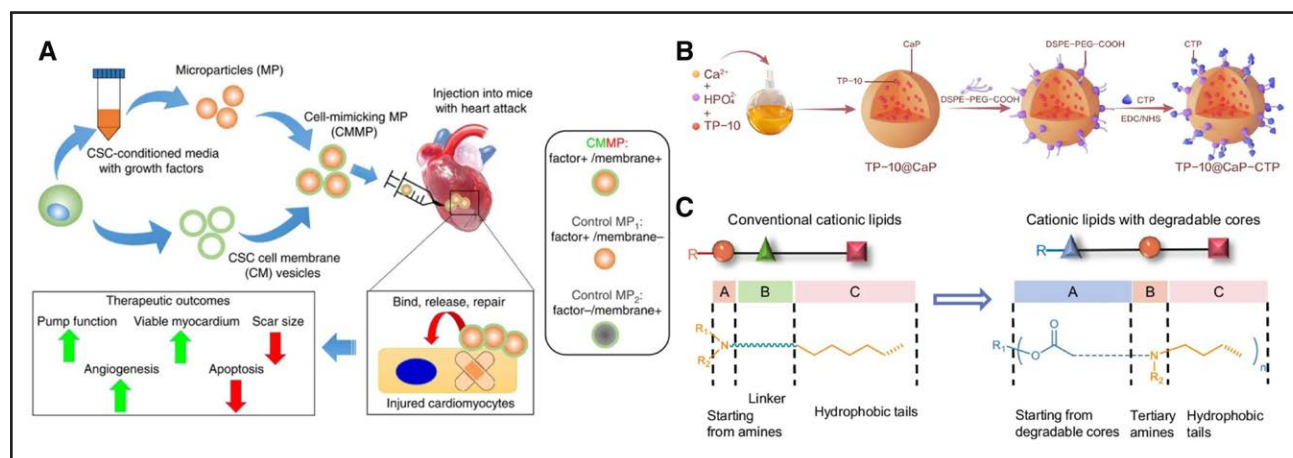


Figure 3. Engineering strategies for targeted delivery of synthetic and biomimetic nanoparticles.

A, Biomimetic cell membrane coating. The synthetic poly(lactic-co-glycolic acid; PLGA) cores are cloaked with native CSC membranes to preserve the surface antigen profiles while providing controlled release. Adapted from Tang et al.⁵⁴ **B**, Extrinsic peptide modification. Nanoparticles are functionalized with CTPs via PEG linkages to better target the heart tissue. Adapted from Weng et al.⁵⁵ **C**, Intrinsic lipid engineering. High-throughput screening and reformulation of the constituent lipids in lipid nanoparticles (LNPs) to achieve organ tropism without the addition of surface ligands. Adapted from Su et al.⁵⁶

strategy has demonstrated targeting to the liver, lung, and spleen, cardiac-specific delivery remains underexplored.⁶¹ Radmond et al reported that the combination of cationic cholesterol and a cationic helper lipid, a dual-charge approach, can achieve mRNA delivery to heart endothelial cells in addition to lung stem cells, providing the first evidence that charge-dependent lipid engineering can reach the myocardium.⁶² This research highlights that cholesterol-based charge modification may have different influences on organ tropism than helper lipid-based modification. Besides organ targeting, researchers have incorporated acid-degradable PEG lipids to enhance diffusion and transfection efficiency. Standard PEG lipids are used for prolonging circulation yet could simultaneously reduce cellular uptake. To solve this limitation, researchers synthesized acid-degradable PEG-LNPs that remain stable in the blood (pH 7.4) but cleave rapidly in the acidic environment of ischemic tissue (pH <6.5).⁶³ This sheddable PEG layer greatly reduces off-target liver uptake and is exposed only after reaching the target site.⁶⁴ Despite these advances, achieving reproducible, efficient cardiac targeting through intrinsic lipid engineering alone remains a challenge. High-throughput combinatorial screening of lipid libraries is essential for optimizing the formulation to achieve high-efficiency cardiac tropism.⁵⁶ Combining intrinsic lipid engineering with extrinsic modification is necessary to overcome the limited accumulation in the heart.

DELIVERY ROUTES

The selection of the administration route is important to overcome rapid cardiac clearance. These delivery routes generally fall into 2 categories: systemic delivery and local delivery. Systemic delivery, such as intravenous injection and intramuscular injection, is the simplest and most standard delivery method.⁶⁵ Nanoparticles are injected into a peripheral vein with the expectation that they will home to the injured area in the heart.^{66,67} This delivery method is less invasive and more translationally relevant. However, intravenous injection shows poor cardiac retention and undergoes rapid clearance by the liver and kidneys. Biodistribution studies have demonstrated that often <1% to 2% of systemically administered nanoparticles reach the targeted cardiac tissue.⁶⁸ As a result, systemic delivery is often coupled with surface modification for better targeting and requires higher doses to achieve therapeutic efficacy, increasing the risk of off-target systemic toxicity.⁶⁹ Inhalation is an emerging systemic delivery strategy that exploits the unique anatomy of pulmonary circulation. Unlike intravenous administration, therapeutics delivered via inhalation enter the pulmonary circulation and can reach the left heart while bypassing hepatic first-pass metabolism. While inhalation was originally used for treating respiratory diseases, recent evidence suggests that the lungs

can serve as a portal for systemic therapy.^{70–72} Specifically, particles such as EVs or nanoparticles (<100 nm) are capable of crossing the air-blood barrier efficiently and accumulating in the injured heart. For example, Li et al developed stem cell-derived exosome nebulization therapy for the treatment of myocardial infarction. They delivered lung spheroid cell-derived secretome/exosomes via inhalation to the heart and demonstrated the capability of these exosomes in reducing fibrotic tissue and enhancing cardiomyocyte proliferation.⁶⁶ This inhalation delivery is noninvasive and needle-free, making it superior for chronic diseases. However, the pulmonary system is evolutionarily designed to exclude particles rather than absorb them. Upon inhalation, the mucus barrier tends to capture the particles, resulting in 90% being cleared away. In addition, long-term exposure to daily inhalation could lead to airway hyperreactivity.⁷³ It is, therefore, important to carefully design the therapeutic vectors to facilitate pulmonary absorption and protect against enzyme degradation. Dosing should also be fine-tuned to ensure therapeutic efficacy while minimizing pulmonary side effects.

Local administration is frequently used for therapeutic delivery in CVD treatment. Intramyocardial injection involves directly injecting the therapeutic agent into the myocardium. This delivery strategy achieves the highest delivery concentration and efficiency at the target area compared with other methods and is favored in animal studies.⁷⁴ Nevertheless, because the heart is a beating pump, the majority of injected therapeutics undergo a washout effect. Researchers have endeavored to incorporate therapeutics into hydrogels for better retention.⁷⁵ Compared with other delivery methods, intramyocardial injection is relatively invasive, requiring a thoracotomy, open-chest surgery, and cardiac exposure, though catheter-based systems are increasing clinical relevance.^{76,77} Intrapericardial injection is a newly developed method for delivering therapeutics to the pericardial cavity.^{78–80} Zhu et al⁷⁹ injected an MSC exosome-loaded hydrogel via intrapericardial injection, which formed a cardiac patch-like structure in the infarct area. The pericardial fluid serves as a depot that allows the drug to diffuse from the outside into the heart muscle, enhancing cardiac retention upon delivery.⁷⁹ Moreover, Zhu et al have demonstrated the pericardial drainage pathway toward the mediastinal lymph node for immune activation upon intrapericardial injection. Following administration, exosomes are preferentially internalized by MHC-II antigen-presenting cells and participate in the PP2A/p-Akt/Foxo3 signaling pathway, which subsequently induces the differentiation and migration of regulatory T cells to modulate inflammation and facilitate long-term heart repair.⁷⁰

To further enhance the therapeutic efficacy of local delivery, biomaterial-based scaffolds have been developed, achieving sustained and controlled drug release while providing mechanical reinforcement to the

weakened ventricular wall. Injectable hydrogels represent one such approach. For example, Hu et al⁸¹ developed an MI-responsive injectable hydrogel encapsulating curcumin-loaded PLGA nanoparticles and recombinant humanized collagen III. They designed this system to degrade on-demand in response to the acidic, ROS-rich infarct microenvironment, releasing anti-inflammatory and proangiogenic payloads to improve cardiac function in a rat MI model.⁸¹ Microneedle patches offer another platform that can be applied epicardially to deliver therapeutics directly into the myocardium. These include cardiac stromal cell-integrated microneedle patches,⁸² detachable microneedle tips that encapsulate MSC-secreted factor-loaded PLGA nanoparticles,¹⁹ and

galunisertib-loaded GalMA microneedle patches for sustained TGF- β inhibition.⁸³ Epicardial patches represent another scaffold strategy, where the patch is applied directly onto the heart surface.⁸⁴ Huang⁸⁵ developed an off-the-shelf artificial cardiac patch composed of a decellularized porcine myocardial extracellular matrix scaffold and synthetic cardiac stromal cells that encapsulate cell-secreted repair factors without requiring live cells. This acellular patch retained its potency after cryopreservation and improved cardiac function in both rat and pig models.⁸⁵ Together, these biomaterial-based platforms illustrate how scaffold design can be tailored to complement diverse delivery routes, offering spatiotemporal control over therapeutic release.

Table. Delivery Routes for Drug Delivery in CVDs

Delivery route	Access method	Target region	Spatial precision	Cardiac retention	Compatible platforms	Advantages	Limitations	Clinical status
Intravenous (systemic) ^{41,57,67,68}	Peripheral vein injection	Systemic delivery relies on passive or active homing to cardiac tissue.	Low, moderate with targeting ligands conjugated or magnetic guidance	Low at the cardiac site for most agents without targeting	MSCs, EVs, nanoparticles, free drugs, antibody-drug conjugates, and LNPs	Least invasive, scalable, no direct heart access required, suitable for systemic anti-inflammatory strategies	Low cardiac retention; most of the cargo ends up in the liver, kidney, or spleen; high dose needed, and difficult to achieve spatiotemporal precision without targeting modifications	Established
Intramyocardial (direct injection) ^{28,43,46,75,81}	Open-chest surgery or catheter-based	Peri-infarct and infarct border zone myocardium	High	Moderate-high (localized depot, but leakage via needle tract)	Hydrogel, stem/progenitor cell, LNP-mRNA, AAV vectors, microparticles, and EVs	Direct delivery to the target tissue, highest local concentration, and compatibility with a sustained and adaptive release system enable precise spatial dosing.	Invasive (open-chest or catheter-based), risk of arrhythmia and perforation, limited injection volume (≈ 0.1 – 0.3 mL per site), and agent washout via needle tract	Emerging/clinical trials
Intracoronary ⁷⁴	Percutaneous catheter via the coronary arteries	Perfused myocardium distal to the infusion site	Moderate	Low-moderate, rapid washout into venous drainage	Stem/progenitor cells, nanoparticles, gene therapy vectors, EVs, and microbubbles	Minimally invasive, integrate with standard PCI workflow, distributes agents across the ischemic area, repeatable, and routes already clinically established	Low retention rate, rapid washout, and limited to the perfused region (not suitable for the no-reflow zone)	Established
Intrapericardial ^{78,79}	Subxiphoid puncture and catheter-based pericardial access	Epicardial surface and outer myocardium via pericardial fluid diffusion	Moderate-high	High within the pericardial cavity and limited transmural penetration	Hydrogels, EVs, gene therapy vectors, nanoparticles, and microrobots/nanorobots	The pericardial cavity acts as a natural drug reservoir and avoids systemic dilution, sustained epicardial exposure, lower immunogenicity risk than IV injection, and access to epicardial vasculature and lymphatics.	Limited transmural drug penetration to endocardial layers, risk of pericarditis, less clinically established, and access is technically demanding.	Experimental/preclinical
Epicardial (patch/scaffold) ^{82,83,85}	Open-chest surgery or minimally invasive thoracoscopy	Epicardial surface directly overlying the infarct zone	High (patch directly placed on the target area)	High, patch/scaffolds remain in situ and achieve sustained local release	Cardiac patches (hydrogel, decellularized ECM, synthetic polymer), cell sheets, and 3-dimensional-bioprinted scaffolds	Good spatial control, sustained and controlled release, and mechanical support to the ventricular wall can deliver cells and therapeutics simultaneously.	Invasive (require surgical access), limited to the epicardial surface, and batch variability of scaffold/patch	Experimental/preclinical
Inhalation ^{71,72}	Nebulized or aerosolized delivery	Pulmonary vasculature to the left heart	Low	Low for cardiac tissue due to pulmonary trapping	EVs, nanoparticles, and inhaled small molecules	Noninvasive, painless, potential for repeated dosing, and avoid first-pass hepatic metabolism	Limited cardiac specificity, primarily explored for pulmonary drug delivery, and cardiac applications are experimental; low delivery efficiency, majority trapped in the pulmonary.	Experimental/preclinical

CVD indicates cardiovascular disease; ECM, extracellular matrix; EV, extracellular vesicle; IV, intravenous; LNP, lipid nanoparticle; MSC, mesenchymal stem cell; and PCI, percutaneous coronary intervention.

TRANSLATION AND FUTURE

In conclusion, nanocarriers have shown superior advantages in delivering therapeutics for CVDs. This review has briefly summarized bioengineered drug delivery systems for CVDs, from conventional synthetic nanocarriers and native EVs to engineered EVs and biomimetic nanoparticles. Each strategy offers distinct advantages; extrinsic surface modifications enhance cell-type specificity, while intrinsic lipid engineering enables scalable organ targeting. In addition, delivery routes and biomaterial scaffolds also play a crucial role in determining how therapeutics reach and are retained in the heart. Collectively, these advances reflect a shift from simple carriers toward integrated, multifunctional platforms that enhance targeting and retention at the injury site.

Despite this progress, several limitations need to be addressed before clinical translation can be realized. First, cardiac targeting efficiency after systemic administration remains low, with typically <1% to 2% of the injected dose reaching the injured site. Extrinsic surface modification and cell membrane coating have improved cardiac accumulation in rodent models, but the absolute gains remain modest. Intrinsic lipid engineering strategies, such as selective organ targeting, have achieved effective delivery to the lungs and spleen, yet a heart-targeted formulation has not been reported. Second, manufacturing scalability and reproducibility can be difficult. Cell membrane-coated nanoparticles and EV-based therapeutics rely on biologically sourced materials, introducing variability in membrane protein composition, vesicle yield, and cargo loading efficiency. In addition, the efficiency of EV production and purification is significantly low. Current isolation methods, including ultracentrifugation, size-exclusion chromatography, and precipitation-based approaches, lack consensus, resulting in batch-to-batch variability in vesicle yield, size distribution, and cargo composition. Scaling these processes to GMP-compliant manufacturing while maintaining consistent potency and sterility remains an open challenge for clinical adoption. Third, immunogenicity concerns persist across multiple platforms. PEGylated nanoparticles can induce the production of anti-PEG antibodies upon repeated dosing, resulting in accelerated blood clearance. For LNP-mRNA systems, ionizable lipids and exogenous RNA can trigger innate immune activation through TLR signaling, and off-target mRNA expression in noncardiac tissues remains a concern for systemically administered formulations. For EV-based therapeutics, the long-term safety profile of membrane-coated particles is largely unknown, and potential cytokine release or procoagulant activity of EV preparations warrants further investigation in repeat-dose studies. These risks vary substantially by platform and should be distinguished between known clinical concerns and those that remain theoretical or preclinical. Among the platforms discussed, LNP-based delivery has

the clearest path to the clinic. The US Food and Drug Administration approval of patisiran and the widespread deployment of mRNA vaccines have already demonstrated the large-scale manufacturing potential of LNP. EVs have a different story: MSC-derived preparations are moving through early-phase trials for heart failure, but the field still lacks agreed-upon manufacturing standards and reliable potency assays. Besides, it is also worth noting that most studies cited in this review rely on rodent models, and validation in large-animal models remains an important step before clinical translation. Nevertheless, the review has centered on myocardial infarction, but the delivery platforms discussed are not really limited to that context. Cardiac hypertrophy, heart failure, inflammatory cardiomyopathy, and arrhythmia all involve variations of the same underlying processes, oxidative stress, persistent inflammation, and fibrosis, which means that the same nanocarrier approaches could plausibly be applied across a much wider range of cardiovascular conditions. As the field develops, broadening the scope seems like a logical progression rather than a stretch.



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Disclosures

None.

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